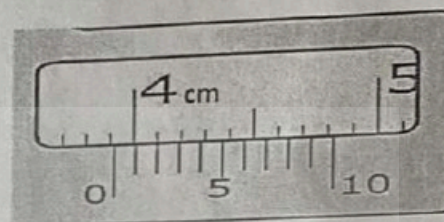
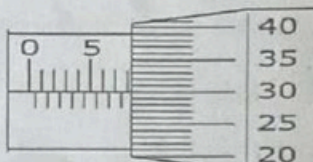
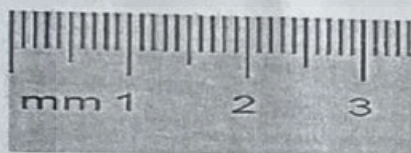


** Circle the correct answer, and move these answer to the table above :(13.5 marks)

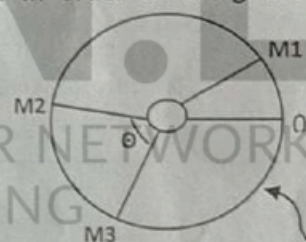
1) Using significant figures rules, find the sum of these readings:



- | | | | |
|----------|----------|-----------|------------|
| A) 51.mm | B) 82.mm | C) 81.5mm | D) 81.50mm |
|----------|----------|-----------|------------|

2) Three forces are not in equilibrium, as $M_1=65g$, $M_2=80g$ and $M_3=50g$. In order to bring the system in equilibrium, the value of θ is:

- | | | | |
|------------------|------------------|------------------|----------------|
| A) 116.7° | B) 125.8° | C) 101.5° | D) 135° |
|------------------|------------------|------------------|----------------|



Force table

3) If a student used an elastic ruler with a maximum range (15 cm) to measure the circumference of a circle which has a radius of (2.2m), then the uncertainty in measuring the circumference is:

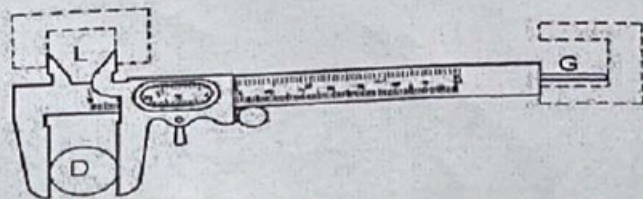
- | | | | |
|------------|------------|-------------|-------------|
| A) 27.6 mm | B) 0.27 mm | C) 22.61 mm | D) 46.05 mm |
|------------|------------|-------------|-------------|

4) Three forces are in equilibrium, $F_{2x} = 0.25 \text{ N}$, $F_{2y} = -0.4 \text{ N}$, the direction of the resultant between F_1 and F_3 (According to positive x-axis) is:

A) 57.99°	B) 122.01°	C) 302.01°	D) 237.99°
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5) In this picture we can say:

A) $G \leq D \leq L$	B) $L = G = D$	C) $L > D > G$	D) $L < D < G$
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6) Using (S.F)-rules. The result of the next addition process ($2.41 + 10.00$) is:

A) 24.1	B) 12	C) 12.41	D) 12.4
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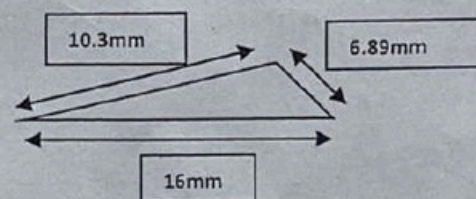
7) Digital balance, when it turned on give us (0.0 g). Then the error in the value of mass using this balance is:

A) 0.1 g	B) 0.01 g	C) 0.005 g	D) 0.05 g
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8) Student used different instruments to calculate the circumference

Of triangle, the error in the circumference is using (S.F) rules:

A) 1mm	B) 0.5 mm	C) 0.56 mm	D) 0.555 mm
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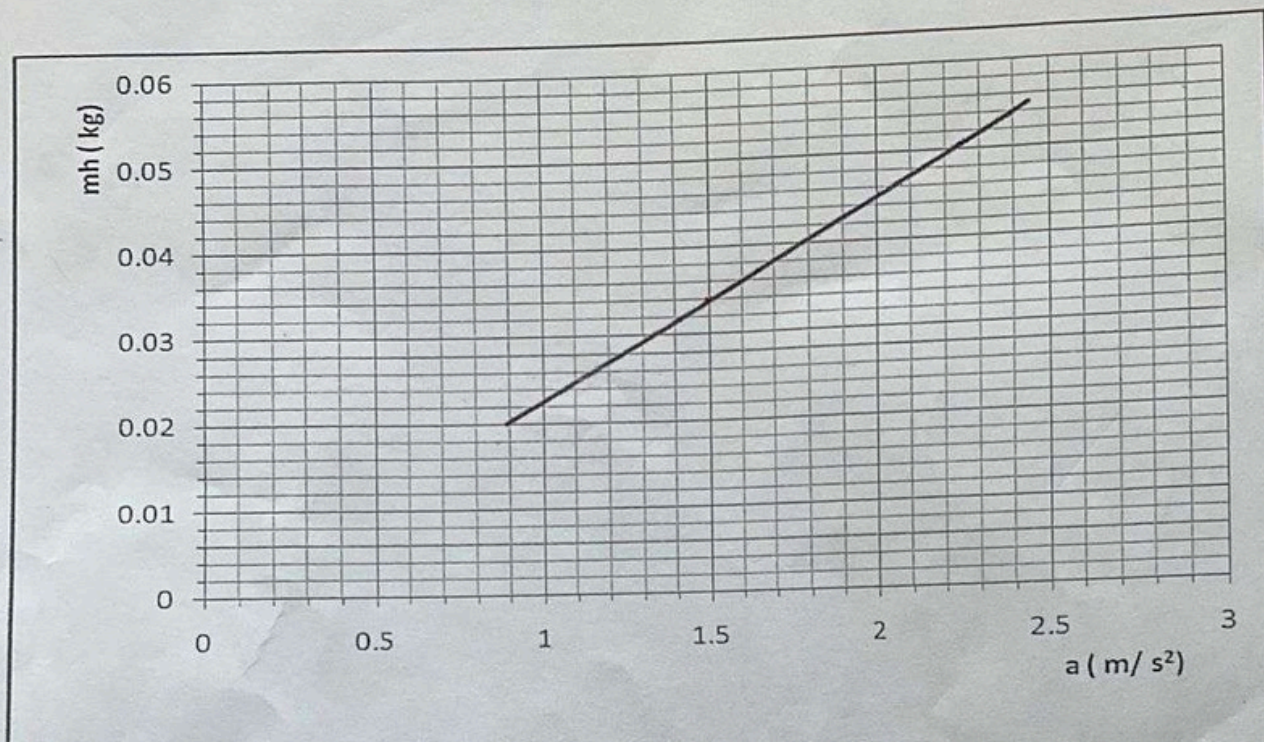


9) A spherical ball with radius (R), the uncertainty (error) in its surface area is:

A) $6A \cdot \Delta R/R$	B) $A \cdot \Delta R/R$	C) $2A \cdot \Delta R/R$	D) $3A \cdot \Delta R/R$
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Q (2): (7.5 marks)

A student drew the graph shown below. In Newton's second law experiment, as (s) and (Mw+mh) are constant in all trials (S=90cm ((where (S) is the distance between the two photo gates)), (Mw+mh=0.480 kg)



Note: Use the last graph only, to answer the next questions.

(1.5 marks) 1) the physical meaning of the slope and find its magnitude:

$$\text{slope} = \frac{M_w + mh}{g} = \frac{0.480}{10} = 0.048$$

$$g = \frac{(M_w + mh)}{\text{slope}} \quad a = \frac{g}{x}$$

(1.5 marks) 2) the experimental value of acceleration due to gravity:

From equation $mhg = (M_w + mh)a \Rightarrow \frac{g}{g} = \frac{mh}{M_w + mh}$ at time = 0.894 = 0.10625

$$a = \frac{2S}{t^2}$$

(1.5 marks) 3) at (t=0.894 s), the force that drives the system:

$$F = (M_w + mh)a = (0.480)(2(0.90)/(0.894)^2) = 1.08 \text{ N}$$

(1.5 marks) 4) the mass of the sliding cart M_w at the same previous time (t=0.894s):

From the graph mh at t=0.894 is mh = 0.051

$$M_w + mh = 0.480$$

$$M_w = 0.480 - 0.051 = 0.429 \text{ kg}$$

(1.5 marks) 5) One aim of Newton's second law experiment is:

Calculate the experimental value of gravity by $F_g = mh \cdot g$ and neglecting all force ~~without~~ like friction and air resistance

Good luck

To study relationship between force and accel when mass is constant